

# Clinical Knowledge Reference - Symptoms, Conditions and Medication Timing

Synthetic enterprise knowledge document for MAOS development. It demonstrates structured clinical knowledge retrieval, exact term matching, hybrid search metadata, and evidence-backed responses. It is not medical advice.

## Cardiometabolic Conditions

| Condition       | Typical symptoms / signals                                                                          | Medication timing guidance                                                                                                                            |
|-----------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hypertension    | Persistent elevated blood pressure; common symptoms can be absent or include headache or dizziness. | Antihypertensive therapy is prescribed by the treating clinician. Timing depends on the prescribed medication and patient factors.                    |
| Type 2 Diabetes | Common symptoms may include increased thirst, frequent urination, fatigue, or blurred vision.       | Diabetes medication timing must follow the clinician-approved regimen; meal-related timing can vary by drug.                                          |
| Hyperlipidemia  | Often asymptomatic; discovered through lipid testing.                                               | Lipid-lowering medication is generally taken according to the prescribed schedule; some agents have timing instructions tied to meals or evening use. |

## Respiratory Conditions

| Condition                                | Typical symptoms / signals                                             | Medication timing guidance                                                                                    |
|------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Asthma                                   | Episodes of wheezing, chest tightness, cough, and shortness of breath. | Controller and rescue therapies have different purposes; medication timing follows the clinician action plan. |
| Community-acquired respiratory infection | May include fever, cough, fatigue, and shortness of breath.            | Antimicrobial therapy, when prescribed, must follow the exact dose and interval supplied by the clinician.    |
| Allergic rhinitis                        | Sneezing, nasal congestion, itchy eyes, and runny nose.                | Antihistamine or nasal therapy timing follows the prescribed instructions; some agents may cause drowsiness.  |

## Gastrointestinal Conditions

| Condition                       | Typical symptoms / signals                               | Medication timing guidance                                                                                  |
|---------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Gastroesophageal reflux disease | Heartburn, regurgitation, or upper abdominal discomfort. | Some acid-suppression therapies are prescribed before meals; exact timing depends on the medication.        |
| Gastritis                       | Upper abdominal discomfort, nausea, or early fullness.   | Medication timing depends on the treatment plan and whether the drug is intended before or after meals.     |
| Irritable bowel syndrome        | Abdominal pain associated with bowel habit changes.      | Treatment can include diet, symptom-directed medication, and clinician follow-up; timing is individualized. |

## Metadata for Retrieval and Authorization

Each document/chunk in the knowledge layer should carry metadata used for routing, filtering, and access control.

| Field          | Example                 | Purpose                             |
|----------------|-------------------------|-------------------------------------|
| document_type  | clinical_reference      | Controls document class             |
| condition_code | HTN-001                 | Exact-code / keyword matching       |
| specialty      | cardiology              | Departmental filtering              |
| access_level   | doctor                  | RBAC filtering                      |
| source_system  | clinical_knowledge_base | Citation/source tracking            |
| version        | 3.2                     | Document version control            |
| effective_from | 2026-01-01              | Policy/knowledge validity           |
| review_status  | approved                | Only approved content is answerable |

## Example retrieval behavior

**User:** "HTN-001 medication timing"

**Retriever:** combine semantic similarity with BM25 keyword matching for the exact code.

**Filter:** specialty=cardiology AND access\_level appropriate for the authenticated role.

**Answer:** return only approved evidence, cite the source document/version, and avoid making an autonomous diagnosis.